



Report

Cryptanalysis

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Substitution Cipher Code

```
import string
```

A list containing all characters #

```
all_letters= string.ascii_letters
```

```
"""
```

create a dictionary to store the substitution

for the given alphabet in the plain text

based on the key

```
"""
```

```
{ } = dict1
```

```
key = 6
```

```
:for i in range(len(all_letters))
```

```
dict1[all_letters[i]] = all_letters[(i+key)%len(all_letters)]
```

```
plain_txt= "Al-Hilal Club is one of the sports clubs in the Kingdom of  
Saudi Arabia. Its main headquarters is based in the capital, Riyadh,  
".and is ranked among eighteen clubs in the Roshn Saudi League
```

```
[]=cipher_txt
```

loop to generate ciphertext #

```
:for char in plain_txt
```

```
:if char in all_letters
```

```
temp = dict1[char]
```

```
cipher_txt.append(temp)
```

```
:else
```

```
temp =char
```

```
cipher_txt.append(temp)
```

```
cipher_txt= "".join(cipher_txt)
```

```
print("Cipher Text is: ",cipher_txt)
```

```
"""
```

create a dictionary to store the substitution

for the given alphabet in the cipher

text based on the key

```
"""
```

```
{ } = dict2
```

```
:for i in range(len(all_letters))
```

```
dict2[all_letters[i]] = all_letters[(i-key)%(len(all_letters))]
```

loop to recover plain text #

```
[] = decrypt_txt
```

```
:for char in cipher_txt
```

```
:if char in all_letters
```

```
temp = dict2[char]
```

```
decrypt_txt.append(temp)
```

```
:else
```

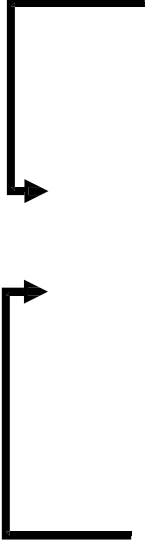
```
temp = char
```

```
decrypt_txt.append(temp)
```

```
decrypt_txt = "".join(decrypt_txt)
```

```
print("Recovered plain text :", decrypt_txt)
```

Output



Cipher Text is: Gr Norg IrAh oy utk ul znk yvuxzy
irAhy ot znk Qotmjus ul YgAjo Gxghog. Ozy sgot
nkgjwAgxzkxy oy hgykj ot znk igvozgr, XoEgjn, gtj oy
xgtqkj gsutm komnzkkt irAhy ot znk Xuynt YgAjo
RkgmAk

.

Recovered plain text : Al Hilal Club is one of the
sports clubs in the Kingdom of Saudi Arabia. Its main
headquarters is based in the capital, Riyadh, and is
ranked among eighteen clubs in the Roshn Saudi
.League

Schedule of Cipher Letters Shifting by 6

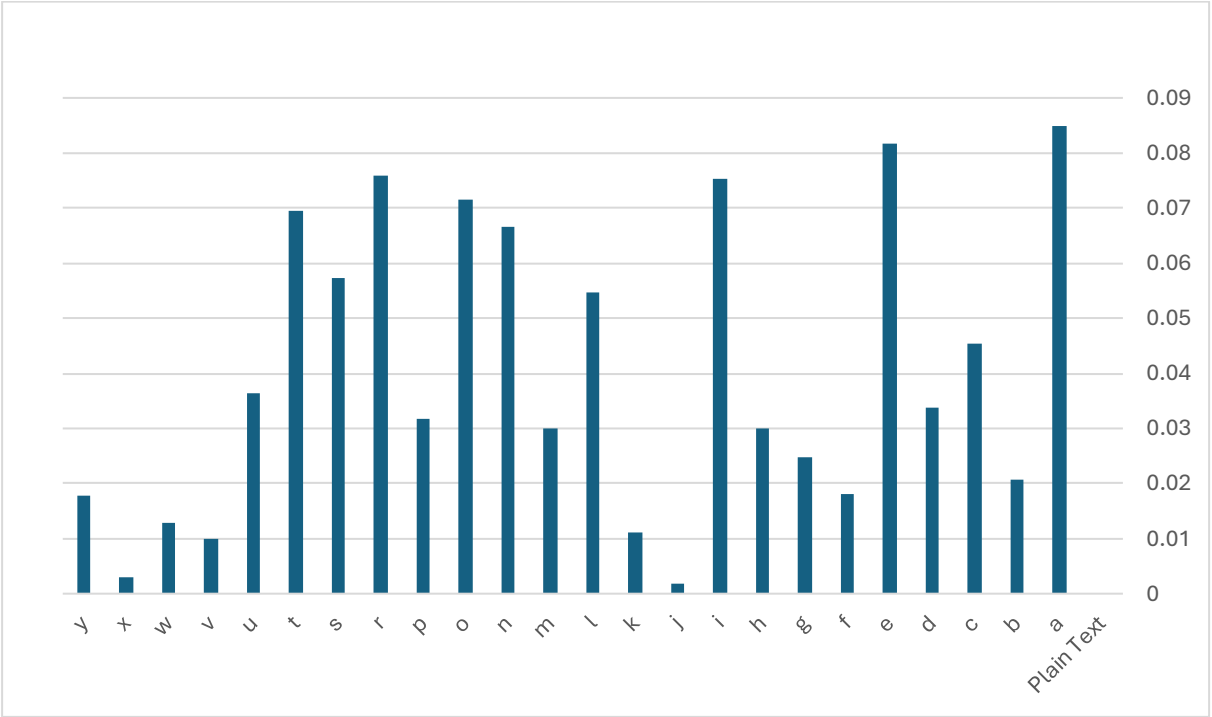
Original Sort of Letters

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f

Letters Shift by 6

Plain Text	Plain Text %	Cipher Text	Cipher Text %
a	8.49%	e	7.19%
b	2.07%	f	1.03%
c	4.53%	g	2.31%
d	3.38%	h	3.59%
e	8.16%	i	8.04%
f	1.81%	j	1.28%
g	2.47%	k	1.80%
h	3%	l	2.05%
i	7.54%	m	8.22%
j	0.19%	n	0.26%
k	1.1%	o	0.51%
l	5.48%	p	3.34%
m	3.01%	q	3.08%
n	6.65%	r	8.22%
o	7.16%	s	4.88%
p	3.16%	t	2.05%
r	7.58%	v	4.62%
s	5.73%	w	5.65%
t	6.95%	x	7.70%
u	3.63%	y	3.08%
v	1%	z	1.28%
w	1.28%	A	1.03%
x	0.29%	B	0.26%
y	1.77%	C	2.31%

Cryptanalysis Before Cipher



Cryptanalysis After Cipher

